# Fact Transaction Analytics Report

## 1. High Level Performance Overview:

- **Total Revenue:** 4.05 billion.
- **Total Profit:** 1,119 million.
- **Profit Margin:** 27.7 %.
- **Total Cost:** 2,929 million.
- **Total Orders:** 60,242
- **Total Products Sold:** 340,176

## 2. Top Performers:

| | | | |
|---|---|---|---|
| 1. | Top Category | Electronics | 3,515 million |
| 2. | Top Product | Laptop | 1,747 million |
| 3. | Top Region | East | 845 million |

| | | | |
|---|---|---|---|
| 1. | Best Performing Month (all years) | September | 417 million |
| 2. | Top Customer Segment | Consumer | 1,362 million |

## 3. Top 10 Product by Revenue:

| Sl. No. | Product Name | Revenue (in million) |
|---|---|---|
| 1. | Laptop | 1,747.4 |
| 2. | Smartphone | 994.4 |
| 3. | Monitor | 559.2 |
| 4. | Desk | 267.2 |

| | | |
|---|---|---|
| 5. | Office Chair | 177.4 |
| 6. | Headphones | 111.9 |
| 7. | Keyboard | 67.9 |
| 8. | Backpack | 63.8 |
| 9. | Mouse | 34.3 |
| 10. | Water Bottle | 17.0 |

## 4. Profitability By Category:

| **Sl. No.** | **Category** | **Profit (in million)** |
|---|---|---|
| 1. | Electronics | 985.9 |
| 2. | Furniture | 110.3 |
| 3. | Accessories | 21.2 |
| 4. | Stationery | 2.1 |

## 5. Regional Sales Performance:

| **Sl. No.** | **Region** | **Sales (in million)** | **Profit (in million)** |
|---|---|---|---|
| 1. | East | 845 | 264.8 |
| 2. | Central | 826 | 244.1 |
| 3. | West | 803 | 209 |
| 4. | South | 791 | 208.4 |
| 5. | North | 781 | 190.9 |

## 6. Monthly Sales (all years – 2024 to 2026):

| Sl. No. | Region | Sales (in million) | Profit (in million) |
|---|---|---|---|
| 1. | January | 363 | 89 |
| 2. | February | 369 | 121 |
| 3. | March | 396 | 116 |
| 4. | April | 352 | 86 |
| 5. | May | 376 | 102 |
| 6. | June | 343 | 83 |
| 7. | July | 367 | 89 |
| 8. | August | 363 | 94 |
| 9. | September | 417 | 158 |
| 10. | October | 241 | 59 |
| 11. | November | 231 | 62 |
| 12. | December | 231 | 59 |

## 7. Sales and Profit Comparison by Channel:

| Sl. No. | Region | Sales (in million) | Profit (in million) |
|---|---|---|---|
| 1. | Distributor | 1,058 | 328 |
| 2. | Marketplace | 1,018 | 295 |
| 3. | Retail Store | 1,011 | 265 |
| 4. | Online | 932 | 224 |
| 5. | Unknown | 29 | 7 |

The above table has a row Unknown because in sales channel there is unavailable data where I don't like to randomly enter any corresponding data. Therefore, I have left them as Unknown.

## 8. Red Flag:

| Sl. No. | Customer ID | Revenue (in million) | Profit (in million) |
|---|---|---|---|
| 1. | CUST-06929 | 67.7 | 67.2 |
| 2. | CUST-09579 | 41.5 | 41.1 |
| 3. | CUST-09682 | 17.8 | 17.5 |
| 4. | CUST-09616 | 15.1 | 14.4 |
| 5. | CUST-05919 | 11.9 | 11.4 |
| 6. | CUST-14734 | 9.7 | 9.2 |
| 7. | CUST-03485 | 8.6 | 8.1 |
| 8. | CUST-14990 | 7.4 | 6.8 |
| 9. | CUST-14467 | 2.3 | 1.8 |
| 10. | CUST- 06616 | 0.74 | 0.7 |

These are red flag sales of about more than 30 customers where there profit percent is very high i.e. more than 60% of the sales
the above table is made of 10 sample customer to highlight how they are in red flag list.

I would like to inform you to review on this particular customer.

# Data Cleaning Brief Report

- Load: Read sales_dataset_training_100k.xlsx, Sheet1, promoted the first row to headers and set the data types
- Drop: remove duplicated data from Order_ID column after reviewing
- Standardize text: Upper cased Order_ID and Customer_ID. Applied proper case to Segment, Region and City. Trimmed spaces on all text columns.
- Round money: Rounded Unit_Price, Sales_Amount, Cost_Amount and Profit to 2 decimals.
- Dates: Added Full_Date as a real date, converted from the Excel serial number in Order_Date.
- Missing and invalid values: Fixed  missing Unit Price, quantity ,discount percent with appropriate parameters.
- Replace value: Replace blank and null text data into Unknown
- Anormaly Flag: Added Is_Anormal = “Yes” when profit is more than 60% of sales.